Marine Biology (2018) 165:17
https://doi.org/10.1007/s00227-017-3273-z

SHORT NOTE

Estimating flight heights of seabirds using optical rangefinders and GPS data loggers: a methodological comparison

Kai Borkenhagen[1] · **Anna-Marie Corman**[1] · **Stefan Garthe**[1]

Received: 20 June 2017 / Accepted: 22 November 2017 / Published online: 4 December 2017
© Springer-Verlag GmbH Germany, part of Springer Nature 2017

Abstract

Determining the collision risk of seabirds with offshore wind farms is crucial for the environmental impact assessment of such installations. The collision risk is often assessed by measuring avian flight heights. Therefore, we measured flight height distributions of 15 seabird taxa, abundant in German offshore waters, with an optical laser rangefinder ($n = 2508$ measurements). For lesser black-backed gulls, we compared these rangefinder measurements to flight heights recorded by GPS data loggers which were attached to 17 birds during incubation. Both methods have specific advantages and disadvantages. Rangefinder measurements are only possible during the day, and rain or fog prevents successful measurements. Data were negatively biased against low and very high flight heights. Since measuring low-flying birds proved more difficult during application of the method in the field, observers should be instructed to dedicate extra effort to measure low-flying birds. Visual observations of low-flying birds can also help to reduce uncertainty and overcome bias problems. Flight heights ranged from – 3 to 431 m above sea level. Most seabirds (70%) flew below rotor level (30 m), but about 30% (mainly large gulls and cormorants) flew in the rotor-swept area. GPS height measurements show a similar general pattern of flight height distribution as rangefinder measurements. However, this method is restricted to few individuals. Both methods complement each other and can provide a reliable estimate of seabirds' flight height distribution.

Introduction

Flight heights of seabirds at sea are an important factor determining collision risks with offshore wind farms. The continuing development of offshore wind farms in European waters and elsewhere thus requires detailed information on flight height profiles for individual birds and flight height distributions for bird species (Furness et al. 2013; Johnston et al. 2014; Cleasby et al. 2015).

The reactions of seabirds to offshore wind farms vary from avoidance and consequent habitat loss (e.g. northern gannets *Morus bassanus*; Garthe et al. 2016) to attraction (e.g. lesser black-backed gulls *Larus fuscus*, herring gulls *Larus argentatus*; Dierschke et al. 2016) and the use of wind farm structures for resting and/or foraging (common gulls *Larus canus*, black-legged kittiwakes *Rissa tridactyla*, great black-backed gulls *Larus marinus*; Vanermen et al. 2015). These latter species are exposed to high risks of collisions with the turbines (reviewed in Dierschke et al. 2016).

Commonly used methods to estimate seabirds' flight heights include radar measurements, visual observations and measurements using optical rangefinders during boat-based surveys, individual flight height data recorded by bird-borne data loggers, and flight height data derived from digital aerial surveys (Camphuysen et al. 2004; Desholm and Kahlert 2005; Dokter et al. 2013; Corman and Garthe 2014; Johnston et al. 2014; Fijn et al. 2015; Johnston and Cook 2016). However, the use of bird-borne data loggers is limited by the inability of most devices to measure height accurately (Cook et al. 2012; Bouten et al. 2013; Corman and Garthe 2014).

Lesser black-backed gulls are migratory seabirds with a manoeuvrable and enduring flight style (Bauer et al. 2005).

Responsible Editor: V.H. Paiva.

Reviewed by J. Miguel Pereira and J. Shamoun-Baranes.

Kai Borkenhagen and Anna-Marie Corman contributed equally.

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s00227-017-3273-z) contains supplementary material, which is available to authorized users.

✉ Kai Borkenhagen
borkenhagen@ftz-west.uni-kiel.de

[1] Research and Technology Centre (FTZ), Kiel University, Hafentörn 1, 25761 Büsum, Germany

They forage in marine and terrestrial habitats and feed on fish, swimming crabs, fisheries discard, earthworms, insects and sometimes garbage (Bauer et al. 2005). They are abundant in German waters and, therefore, well suited as a model organism to compare different measurement methods of avian flight heights.

The objectives of this study are to present flight height data for several common seabird species, and to compare data generated using optical laser rangefinders with data generated by GPS loggers, as well as to assess the advantages and disadvantages of these techniques. We additionally give recommendations regarding the future uses of both methodologies.

Materials and methods

Rangefinders

A hand-held Vector 21 Aero laser rangefinder (Vectronix AG, Switzerland) was used to record flight heights of each bird during ship-based seabird-at-sea transect surveys. A total of 2508 measurements were recorded in the time period between February 2010 and January 2015, 90.8% in the German part of the North Sea and 9.2% in the German part of the Baltic Sea. Measurements were not restricted to offshore wind farm areas. 14.2% of the measurements were taken in winter (December–February), 18.7% in spring (March–May), 64.7% in summer (June–August) and 2.4% in autumn (September–November). All measurements were taken during daylight hours and during dry conditions, because fog and rain interfere with rangefinder measurements. High sea states lead to increased movement of the survey vessel and impede measurements at large distance. During favourable conditions, birds of the size of lesser black-backed gulls were successfully measured up to distances of around 500 m. To increase measurement reliability, each bird was measured at least twice (one measurement following immediately after the other), and flight heights were only recorded if two measurements differed by ≤ 5 m. This prevented measurements of birds that changed their flight height rapidly. A rangefinder determines the height above or below the observer's eye level, and flight height above sea level (asl) was, therefore, obtained by adding the deck height and eye height to each measurement, individualized for each vessel and observer.

Measurements of flight heights obtained using laser rangefinders under-represent low-flying birds, because it is harder to get successful measurements for these. Flight heights for low-flying birds (2 m—sea level) were, therefore, estimated visually and assigned to 1 m asl.

According to the manufacturer's data the accuracy of rangefinder measurements is about 2 m. We had no opportunity to check this claim, but field experience suggests that measurement errors occur and thus we repeated each measurement (see above).

GPS loggers

Individual flight height data (flight positions: all global positioning system (GPS) logger geofixed with an instantaneous speed > 10 km h^{-1}, $n = 2714$; Shamoun-Baranes and van Loon 2006; Klaassen et al. 2011; Corman and Garthe 2014) were obtained from tracking studies of 17 lesser black-backed gulls equipped at three different breeding colonies (Helgoland: $n = 6$, Juist: $n = 6$, Norderney: $n = 5$) in the North Sea during incubation in the mid of May 2013 (data were taken from Corman and Garthe 2014) and 2014. Capturing and equipment procedures are explained by Corman and Garthe (2014).

Flight height measurements are generally difficult to obtain using most of GPS logger systems and the results are characterized by their uncertainty (Cook et al. 2012; Bouten et al. 2013). The GPS loggers used in the current study (46.5 mm × 32 mm × 18.5 mm; 23 g; earth and Ocean Technologies, Kiel, Germany) were, therefore, specially programmed to ensure more accurate height measurements with an accuracy of about ± 10 m: Flight heights were recorded continuously (i.e. every second) for 35 s, at 5-min intervals, 24 h a day until the battery was empty (resulting in a relatively short running time of the devices of about 2–4 days; details are explained in Corman and Garthe 2014).

Data analysis

Data were explored with simple summary statistics and compared in diagrams. No statistical tests were performed. For the analysis of height data from GPS loggers, we took the last (i.e. the 35th) continuously measured value at every 5 min (resulting in one measurement for each 5-min interval). Each measurement (rangefinder and GPS logger) was treated as independent measurement, although height data from GPS loggers might be spatially and temporally autocorrelated depending on the sampling interval. The dataset is presented in the supplementary file.

Results and discussion

Estimated flight height distributions and comparison of measurement methods

A total of 2508 flight height measurements were obtained from 15 common seabird species or species groups using optical laser rangefinders, plus an additional 280 visual estimates of low-flying birds (< 2 m asl). Flight heights ranged

from − 3 m to 431 m asl. Most seabirds (70%), including wind-farm-sensitive scoters, auks, gannets and divers (Garthe and Hüppop 2004; Dierschke et al. 2016), flew below 30 m asl and thus below the critical rotor-swept area of most turbines (30–150 m asl; Figs. 1, 2). However, about 30% of the measured birds (mainly large gulls and to some extent cormorants, *Phalacrocorax carbo*) flew within the rotor-swept area (Fig. 1, Table 1). The flight height distribution during foraging flights of the 17 lesser black-backed gulls ($n = 735$ recorded positions in flight) equipped with GPS loggers revealed similar results; birds mostly flew at heights between 0 and 30 m (Fig. 2, Table 1), while about 17% of the recorded heights were within the rotor-swept area of the wind farms (Table 1).

These results are largely in line with the findings of several impact assessment studies (mostly based on boat-based survey data) at different offshore wind farms in the North and Baltic Seas (reviewed in Dierschke et al. 2016). Large gulls and cormorants are known to be attracted to offshore wind farms, and are thus at particularly high risk of colliding with the turbines (Garthe and Hüppop 2004; Vanermen et al. 2015; Dierschke et al. 2016). Only herring gulls and lesser black-backed gulls occasionally flew above 150 m [0.4% of heights measured by rangefinders; 19% of heights recorded by GPS loggers, while 0.3% ($n = 7$) of the recordings were ≤ − 10 m; Table 1]. Two GPS measurements (− 67 and − 56 m asl) were presumed to be measurement errors and were removed from the dataset.

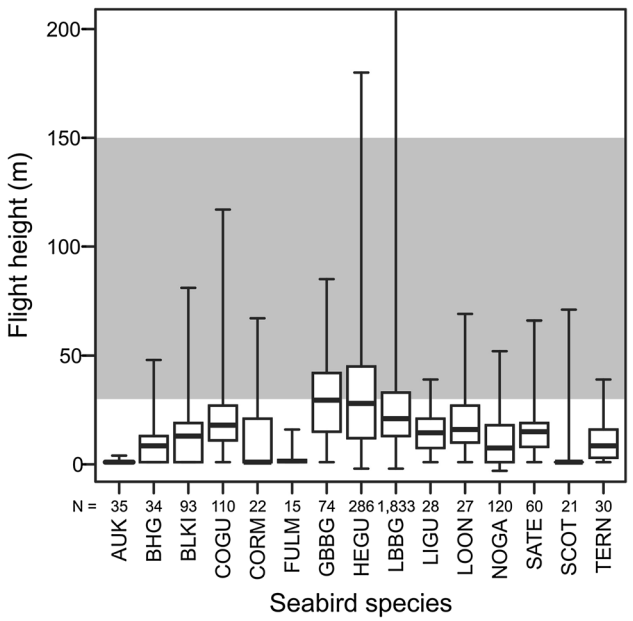


Fig. 1 Flight heights of common seabird species in relation to the typical rotor-swept area of offshore wind turbines based on rangefinder measurements and visual estimates during boat-based surveys. *AUK* razorbill, guillemot, unidentified auk, *BHG* black-headed gull, *BLKI* black-legged kittiwake, *COGU* common gull, *CORM* great cormorant, *FULM* northern fulmar, *GBBG* great black-backed gull, *HEGU* herring gull, *LBBG* lesser black-backed gull, *LIGU* little gull, *LOON* red- and black-throated divers, unidentified divers, *NOGA* northern gannet, *SATE* sandwich tern, *SCOT* common scoter, *TERN* arctic and common terns, unidentified terns. The box represents the interquartile range; the bold horizontal bar represents the median. The whiskers are drawn from the first or third quartile to the lowest or largest value within 1.5 times the interquartile range, respectively

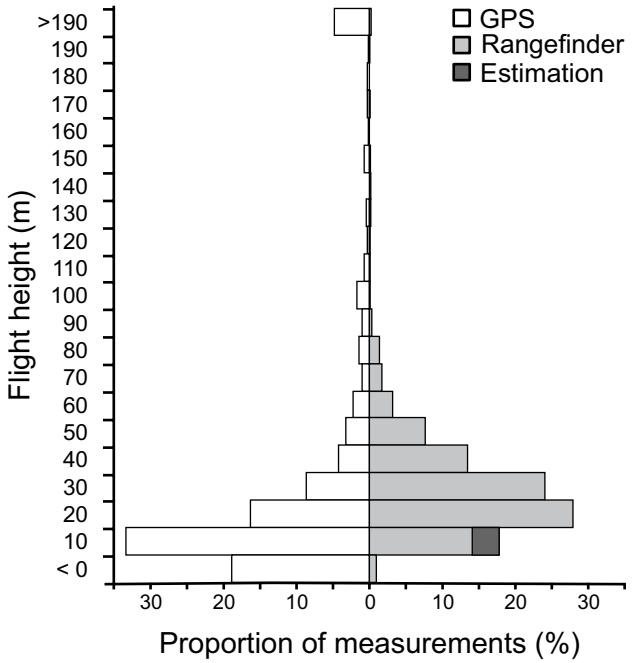


Fig. 2 Comparison of flight height distributions of lesser black-backed gulls measured by rangefinder and GPS loggers. Values on the *y*-axis represent the upper bound of the corresponding class. The right part of the figure also includes the visual estimates of low-flying birds

Negative heights most likely represent times when birds were foraging near the water surface.

Both methodologies can provide a picture of the flight height distribution of seabirds at sea. However, both techniques have advantages and disadvantages (Table 2). A laser rangefinder can be used to record one single flight height value per bird and measurements can be repeated for multiple individuals. In contrast, GPS loggers record repeated flight heights for a few tagged individuals with a given interval and over a given time period, which could be of particular interest for estimating collision risks of seabirds flying within or near offshore wind farms (Cook et al. 2012; Johnston et al. 2014). Furthermore, unlike rangefinders, data acquisition from loggers is not restricted by light or weather conditions. However, the major disadvantage of the devices used in this study was the short battery life (about 2–4 days), owing to the necessary recording schedule for accurate flight height measurements (see above and Corman and Garthe 2014). This disadvantage might be irrelevant when using

Table 1 Flight height distribution of seabird species with > 20 rangefinder measurements

	Rangefinder											GPS
	NOGA	SATE	BLKI	TERN	LBBG	BHG	GBBG	HEGU	LOON	COGU	LIGU	LBBG
n	79	49	68	28	1785	22	67	233	23	105	25	705
Max	52	66	81	39	431	47	85	180	69	117	39	735
Median	14	17	16	9	21	12	32	31	19	18	18	8
Min	− 3	4	0	1	− 2	4	5	− 2	4	2	2	− 10
% 0–30 m	87.3	95.9	83.8	96.4	70.0	95.5	44.8	45.5	87.0	82.9	96.0	59.3
% > 30–150 m	12.7	4.1	16.2	3.6	29.6	4.5	55.2	54.1	13.0	17.1	4.0	17.0
% > 150 m	0	0	0	0	0.4	0	0	0.4	0	0	0	5.7

Heights of 30–150 m represent the typical rotor-swept height range of offshore wind turbines. GPS data: percentages do not sum to 100%, because the remaining 19% of the data were below 0 m. For abbreviations of species names see Fig. 1

Table 2 Comparison of measurement methods for avian flight heights, optical laser rangefinder vs. GPS logger

		Optical laser rangefinder		GPS data logger	
Logistic requirements		Boat-based surveys		Permits for catching, handling and equipping the birds	
Application/implementation		Easy to apply even during routine monitoring surveys	+	For specific study questions	
Bird species/size		Large number of species can be measured; only limited by observer skills (more difficult to aim at small birds)	+	Mostly limited to medium to large birds, depending on size and mass of device	−
Sample size		Moderate; depends on survey duration and number of observers	+	Generally low, but depends on study details	−
Sampling strategy		One value per bird for many different individuals → Only a snapshot of a flying individual		Repeated measurements of few birds, duration depends on type of device → More realistic flight distribution of the relevant individual	+
Accuracy		~ 2 m (according to the manufacturer's data)		<10 m; might be even more accurate with newer types of GPS loggers and options such as 3D accelerometers or pressure sensors	
Autocorrelation of data		No autocorrelation	+	Depends on sampling interval (increases with shorter interval); here, no autocorrelation was found	
Influencing factors	Time of day	Limited to daylight hours	−	No limitations	
	Season	No limitations	+	Here, limited to incubation, but this disadvantage becomes largely irrelevant for solar-powered devices, which can be reprogrammed remotely	(−)+
	Weather	Limited by fog, rain and strong winds	−	No limitations	+
	Location	Large areas can be covered according to the relevant study	+	Mostly limited to the home range of the equipped species	
Impact on birds		Non-invasive	+	Invasive	−

+ = advantages, − = disadvantages; if nothing is indicated, there is no obvious pro or con

the modern solar-powered devices allowing for adjusting the recording schedule remotely (Ross-Smith et al. 2016). The use of GPS devices is restricted to species that are large and heavy enough to allow their attachment, and permits are required for catching, handling and equipping the birds. These limitations suggest that data loggers may not be suitable for measuring flight heights just by themselves. Instead of their standalone deployment, these devices are typically employed as part of a broader project framework with specific questions (Ross-Smith et al. 2016). Furthermore, the accuracy and reliability of the flight height data returned by GPS loggers were approximately ± 10 m (for

details see Corman and Garthe 2014), which was lower than the manufacturer's data for the rangefinder of about 2 m. In contrast, rangefinders are non-invasive and can be applied easily to a wide range of species at the same time. However, for rangefinder measurements, boat-based seabird surveys have to be funded and organized. GPS measurements might be influenced by effects of the device on bird's behaviour, while rangefinder measurements might be influenced by the reaction of the bird to the approaching survey vessel. To minimize the impact of these factors, we employed GPS devices not weighing more than 5% of birds' averaged body mass (Corman and Garthe 2014), monitored the birds after equipment intensively, and excluded birds that reacted to the survey vessel from the analysis of the rangefinder measurements.

Despite the general advantages and disadvantages of both methods, our results showed similar flight height distributions for lesser black-backed gulls measured by both device types. The main difference is the wider range of flight heights recorded by the data loggers compared with those measured by the rangefinder. Furthermore, the proportion of height measurements within the rotor-swept area according to the rangefinder data was nearly twice as high as that based on the logger data. Both measurement methods provided relatively high proportions of negative heights, probably due to presumed foraging flights close to the sea surface. Because GPS loggers are not restricted by weather, light conditions and time of day, they are more likely to provide a random sample of flight heights and thus a less biased representation of the true flight height distribution. In addition, rangefinder measurements are biased against very low and very high flight heights (Desholm et al. 2006; Kahlert et al. 2012; Johnston and Cook 2016) compared with GPS data. Although it is generally possible to measure very low flying birds (< 10 m asl) with a rangefinder, field experience shows that it is difficult in practice. We attempted to compensate for this rangefinder bias using visual observations, although these are typically not very precise (Camphuysen et al. 2004; Johnston and Cook 2016). To obtain reliable estimates, we, therefore, only estimated the flight heights of birds flying below 2 m asl (i.e. directly above the water surface).

Ross-Smith et al. (2016) provided flight height distributions of lesser black-backed gulls and great skuas *Stercorarius skua* modelled from flight heights measured by GPS loggers. Although they categorized the height measurements according to location (terrestrial, coastal, marine) and time of day (during the day, during twilight, at night), which complicates the direct comparison, the modelled proportions of birds at risk height were roughly in line with our findings. Johnston and Cook (2016) modelled the proportion of birds flying between 20 and 120 m using boat-based and digital aerial survey data to estimate the collision risks of seven seabird species. Their results clearly differed from our measurements using rangefinders. We found much higher proportions of birds within this altitude range for kittiwakes, herring gulls, and lesser and great black-backed gulls (Table 3). Our rangefinder measurements for sandwich terns *Thalasseus sandvicensis* and northern gannets were higher than their boat-based estimates, but lower than flight height measurements from digital aerial surveys (Table 3). Digital aerial survey methods (e.g. APEM, http://www.apemltd.co.uk/service/aerial-surveys-bird-marine-mammal/; DAISI, http://www.ifaoe.de/en/services/ornithology/daisi.html; HiDef, Weiß et al. 2016) are used increasingly to acquire flight height data for seabirds. They are based either on the bird's parallax in front of the background (HiDef) or estimated by the bird's size in the image compared with average absolute bird sizes from a database (APEM, DAISI). The latter method in particular relies on estimates from images and mean values from the literature, leading to uncertainties related to image quality, variation in body size, and bird behaviour. These uncertainties reduce the accuracy of digital aerial survey methods (Johnston and Cook 2016) and might explain the apparent differences compared with other methods.

Recommendations and future prospects

The results of the present study suggest that rangefinder data are negatively biased against low and very high flight heights. Observers should be instructed to dedicate extra effort to measuring low-flying birds, while the implementation of visual estimates of low-flying birds can also help to alleviate this bias.

The recent progress in bio-logging technologies, with accurate, smaller and increasing battery life of tracking

Table 3 Proportions of bird observations between 20 and 120 m asl, the rotor-swept area use by Johnston and Cook (2016), measured using different methods

	BLKI (%)	LBBG (%)	GBBG (%)	HEGU (%)	NOGA (%)	SATE (%)
a) Boat based[a]	15	28	33	32	13	7
b) Digital aerial survey[a]	13	42	26	36	50	57
Laser rangefinder	32.4	53.7	74.6	79	30.4	26.5
GPS logger	–	24.5	–	–	–	–

For abbreviations of species names see Fig. 1

[a]Data from Johnston and Cook (2016)

devices, allows to increase the reliability of flight height data. In addition, triaxial accelerometers are implemented in most new devices, which allow the identification of different types of behaviour (e.g. foraging, commuting) and provide the opportunity to differentiate between flight heights according to specific bird behaviours (Cleasby et al. 2015). It is also possible to combine flight height data from rangefinders with behavioural observations, but behavioural data have not been recorded systematically to date. Additional tracking studies should be carried out using durable/long-lasting GPS tracking devices appropriate for the accurate measurement of flight heights. Furthermore, special attention should be paid to species that are attracted to wind farms, such as cormorants and large gulls, because these are at particular risk of colliding with the turbines (Dierschke et al. 2016).

Flight height measurements acquired using GPS devices and laser rangefinders can be used to develop collision risk models, as they record individual continuous data or are easy to apply to standard boat-based surveys, respectively. However, neither method can provide the same high amount of data, as can be provided by methods like radars or digital aerial surveys. So, both methods are well suited for small-scale studies, e.g. for specific locations and/or species and for specific questions, which require a certain accuracy of flight heights, but can also be used in addition to large-scale studies to assess the impact of offshore wind farms on seabirds (Ross-Smith et al. 2016). Regardless of the measurement method, avian flight height distributions can be influenced by several factors such as weather, time of day, season or location (Shamoun-Baranes et al. 2006; Thaxter et al. 2015; Johnston and Cook 2016; Ross-Smith et al. 2016). That is why comparative studies employing as many different measurement methods and covering as many different environmental conditions as possible are urgently needed, not only to complete and improve collision risk assessments of several seabird species but also to evaluate and calibrate these methods against each other.

Acknowledgements We thank J. Kotzerka, B. Mendel, J. Sommerfeld and V. Corman for help with data logger fieldwork, and all observers who measured flight heights during boat-based SAS surveys. N. Markones managed the survey database and provided help with database queries. We also thank the National Park Administration of the Wadden Sea National Park of Lower Saxony (Germany), and J. Dierschke and the Institute of Avian Research on Helgoland/Wilhelmshaven (Germany) for logistic support during fieldwork. G. Peters gave technical support with the data loggers. S. Furness provided linguistic support. Major parts of the fieldwork were funded within the frameworks of the projects TESTBIRD (Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety; Grant no. 0327689A/FTZ3), WINDBIRD (Federal Ministry for Economic Affairs and Energy on the basis of a decision by the German Bundestag; Grant no. 0325281) and the Marine Biodiversity Monitoring Programme of the German Federal Agency for Nature Conservation (BfN).

Compliance with ethical standards

Conflict of Interest The authors declare that they have no conflict of interest.

Ethical approval All applicable national and institutional guidelines for the capture and handling of animals were followed. All birds were caught, ringed, and equipped with data loggers under licenses issued by the State Agency for Agriculture, Environment and Rural Areas Schleswig–Holstein and the Ministry of Energy transition, Agriculture, Environment and Rural Areas Schleswig–Holstein, Germany (file numbers: V 312-72241.121-37 (34-4/11), V 312-7224.121-37 (80-6/13)), the National Park Administration of the Wadden Sea National Park of Lower Saxony, and the Lower Saxony State Office for Consumer Protection and Food Safety (file number: 33.14-42502-04-11/0666). All birds were handled in strict accordance with good animal practice. Handling time of all birds was reduced to a minimum to minimize stress during capture.

References

Bauer H-G, Bezzel E, Fiedler W (2005) Das Kompendium der Vögel Mitteleuropas: Nonpasseriformes—Nichtsperlingsvögel, 2nd edn. Aula, Wiebelsheim

Bouten W, Baaij EW, Shamoun-Baranes J, Camphuysen CJ (2013) A flexible GPS tracking system for studying bird behaviour at multiple scales. J Ornithol 154:571–580. https://doi.org/10.1007/s10336-012-0908-1

Camphuysen KJ, Fox AD, Leopold MF, Petersen IK (2004) Towards standardised seabirds at sea census techniques in connection with environmental impact assessments for offshore wind farms in the U.K.: a comparison of ship and aerial sampling methods for marine birds, and their applicability to offshore wind farm assessments (No. BAM—02-2002). NIOZ Commissioned by Cowie Ltd

Cleasby IR, Wakefield ED, Bearhop S, Bodey TW, Votier SC, Hamer KC, Österblom H (2015) Three-dimensional tracking of a wide-ranging marine predator: flight heights and vulnerability to offshore wind farms. J Appl Ecol 52:1474–1482. https://doi.org/10.1111/1365-2664.12529

Cook ASCP, Johnston A, Wright LJ, Burton NHK (2012) A review of flight heights and avoidance rates of birds in relation to offshore wind farms. Strategic Ornithological Support Services SOSS-02. BTO Research Report No. 618. BTO, Thetford

Corman A, Garthe S (2014) What flight heights tell us about foraging and potential conflicts with wind farms: a case study in Lesser Black-backed Gulls (*Larus fuscus*). J Ornithol 155:1037–1043

Desholm M, Kahlert J (2005) Avian collision risk at an offshore wind farm. Biol Lett 1:296

Desholm M, Fox AD, Beasley PDL, Kahlert J (2006) Remote techniques for counting and estimating the number of bird-wind turbine collisions at sea: a review. Ibis 148(s1):76–89. https://doi.org/10.1111/j.1474-919X.2006.00509.x

Dierschke V, Furness RW, Garthe S (2016) Seabirds and offshore wind farms in European waters: avoidance and attraction. Biol Cons 202:59–68

Dokter AM, Baptist MJ, Ens BJ, Krijgsveld KL, van Loon EE (2013) Bird radar validation in the field by time-referencing line-transect surveys. PLoS ONE 8(9):e74129

Fijn RC, Krijgsveld K, Poot MJM, Dirksen S (2015) Bird movements at rotor heights measured continuously with vertical radar at a Dutch offshore wind farm. Ibis 157(3):558–566

Furness RW, Wade HM, Masden EA (2013) Assessing vulnerability of marine bird populations to offshore wind farms. J Environ Manage 119:56–66

Garthe S, Hüppop O (2004) Scaling possible adverse effects of marine wind farms on seabirds: developing and applying a vulnerability index. J Appl Ecol 41:724–734

Garthe S, Markones N, Corman A-M (2016) Possible impacts of offshore wind farms on seabirds: a pilot study in Northern Gannets in the southern North Sea. J Ornithol 158:345–349. https://doi.org/10.1007/s10336-016-1402-y

Johnston A, Cook ASCP (2016) How high do birds fly? Development of methods and analysis of digital aerial data of seabird flight heights. BTO Research Report No. 676. BTO, Thetford

Johnston A, Cook ASCP, Wright LJ, Humphreys EM, Burton NHK (2014) Modelling flight heights of marine birds to more accurately assess collision risk with offshore wind turbines. J Appl Ecol 51:31–41

Kahlert JA, Leito A, Laubek B, Luigujoe L, Kuresoo A, Aaen K, Luud A (2012) Factors affecting the flight altitude of migrating waterbirds in Western Estonia. Ornis Fenn 89:241–253

Klaassen RHG, Ens BJ, Shamoun-Baranes J, Exo K-M, Bairlein F (2011) Migration strategy of a flight generalist, the Lesser Black-backed Gull *Larus fuscus*. Behav Ecol 23(1):58–68

Ross-Smith VH, Thaxter CB, Masden EA, Shamoun-Baranes J, Burton NHK, Wright LJ, Rehfisch MM, Johnston A (2016) Modelling flight heights of Lesser Black-backed Gulls and Great Skuas from GPS: a Bayesian approach. J Appl Ecol 53:1676–1685. https://doi.org/10.1111/1365-2664.12760

Shamoun-Baranes J, van Loon E (2006) Energetic influence on gull flight strategy selection. J Exp Biol 209:3489–3498

Shamoun-Baranes J, van Loon E, van Gasteren H, van Belle J, Bouten W, Buurma L (2006) A comparative analysis of the influence of weather on the flight altitudes of birds.Bull Am Meteorol Soc 87(1):47–61

Thaxter CB, Ross-Smith VH, Bouten W, Clark NA, Conway GJ, Rehfisch MM, Burton NHK (2015) Seabird-wind farm interactions during the breeding season vary within and between years: the case study of Lesser Black-backed Gull *Larus fuscus* in the UK. Biol Conserv 186:347–358. https://doi.org/10.1016/j.biocon.2015.03.027

Vanermen N, Onkelinx T, Courtens W, Van de Walle M, Verstraete H, Stienen EWM (2015) Seabird avoidance and attraction at an offshore wind farm in the Belgian part of the North Sea. Hydrobiologia 756:51–61

Weiß F, Büttger H, Baer J, Welcker J, Nehls G (2016) Erfassung von Seevögeln und Meeressäugetieren mit dem HiDef Kamerasystem aus der Luft. Seevögel 37(2):14–21